circuit-board computer and had a 16-bit word-length and provided computation and electronic interfaces for guidance, navigation, and control of the spacecraft. At the time of the project's inception, the AGC was considered to be the riskiest item in the Apollo Project as it was the initial stage of the use of the monolithic integrated circuits which reduced size and weight of the total apparatus.

F-14 Tomcat

The F-14 "Tomcat"

The Central Air Data Computer (CADC) built into the F-14A Tomcat and launched in 1970 was the first microprocessor-based embedded system. It provided cockpit information such as altitude and air-speed and also handled avionics and flight handling computation of the Tomcat's swing-wing configuration. The CADC consists of 8 microprocessors, 19 memory chips, and was designed by MOS-LSI. For guiding the aircraft control surface no cams or gears were used – everything was controlled by hydraulic systems and electric motors and so the CADC represented the first completely digital control system for guiding aircraft control surfaces. Details of the CADC only came out in 1998 when its lead designer published a paper. We think it's safe to assume that the lead designer lived a long, troubled life, being stalked night and day by armed, black-suit-wearing men called Agent Smith, who work for the "Government"'.

Modern usage

As you can see from the examples above, military and aviation applications drove embedded systems during its initial stages. Thankfully, from 1990 onwards things have changed for the better. The leading industries reaping the benefits of embedded systems have been the healthcare industry with integrated diagnostic devices, progressing to cellphones. The driving force behind most of these new applications of embedded systems is the availability of 32-bit processors and the fact that development tools are easier to use. These, in turn, have increased the usable computational power